

MJ.23.R2 is a risk detection and governance framework designed to identify patterns of inaccurate reporting, settlement distortion, behavioral inconsistency, and post-settlement re-exposure. By integrating variance analytics, attribution models, settlement evaluation, escalation mechanisms, and governance scoring, the framework provides an evidence-based approach for prioritizing review of situations that may increase operational, financial, compliance, or legal liability.

The framework applies layered scoring engines to identify operational and financial anomalies:

$$\text{Variance\%} = |(\text{Reported} - \text{Verified}) / \text{Verified}| \times 100$$

$$\text{Incidence} = (\text{Incorrect} / \text{Total}) \times 100$$

$$\text{Consistency} = (\text{Successful} / \text{Total}) \times 100$$

These core signals feed into higher-order models:

$$\text{SVP} = 0.30(\text{Variance}) + 0.20(\text{Trend}) + 0.30(\text{Incidence}) + 0.20(100 - \text{Consistency})$$

$$\text{Risk} = 0.40(\text{Incidence}) + 0.25(\text{Variance}) + 0.20(\text{Exceptions}) + 0.15(100 - \text{Accuracy})$$

$$\text{Settlement Risk} = 0.40(100 - \text{Efficiency}) + 0.35(\text{Backlog}) + 0.25(100 - \text{Achievement})$$

$$\begin{aligned} \text{FII} = & 0.25(\text{Variance}) + 0.20(100 - \text{Liability}) + 0.15(\text{Doubt}) + 0.20(\text{Negotiator Risk}) \\ & + 0.20(\text{Settlement Risk}) \end{aligned}$$

The framework also incorporates escalation logic to reduce false positives while identifying multi-factor risk convergence:

Double Trigger: $\text{Variance} \geq 60$ AND $\text{Incidence} \geq 60$ OR Settlement Variance Condition

Triple Trigger: $\text{Variance} \geq 60$ AND $\text{Incidence} \geq 60$ AND Legal Action Exists

While conceptual and non-production, MJ.23.R2 demonstrates my ability to translate operational experience into structured analytical systems using basic R logic and statistical modeling principles.

My objective is to apply this approach to help identify early indicators of settlement inefficiency, behavioral variance, compliance risk, and emerging operational trends before they escalate into material issues.

MJ.23.R2
“Beyond Integrity”

Financial Reporting Integrity, Settlement Variance, Fraud and Default Risk Calculator, Reasonable Doubt Engine and Negotiator Attribution Framework

Intended to provide operators easy and early Detection of Reporting Distortion, Settlement Misvaluation, and Legal Exposure Risk

Summary

MJ.23.R2 is a layered financial risk analytics framework designed to identify patterns of reporting variance, recurring inaccuracies, settlement misvaluation, behavioral inconsistency, and post-settlement re-exposure risk within debt settlement operations.

The framework evaluates reported financial information against verified financial information and applies a series of statistical scoring engines to detect:

- Persistent reporting inaccuracies
- Repeated settlement misvaluation
- Negotiator-attributable variance patterns
- Settlement process degradation
- Escalating creditor dispute exposure
- Post-settlement failure events
- Increased legal and governance risk

The objective is to identify conditions that may increase liability exposure for:

1. The Client
2. The Debt Settlement Company
3. Creditors
4. Negotiators

SECTION 1 — CORE MATHEMATICAL FOUNDATION

1.1 Variance Percentage Engine

Purpose

Measures deviation between reported and verified financial values.

Equation

$$\text{Variance\%} = |(\text{Reported} - \text{Verified}) / \text{Verified}| \times 100$$

R Implementation

```
variance_engine <- function(reported, verified){ if(verified <= 0) return(0) abs((reported -  
verified) / verified) * 100  
}
```

Interpretation

Variance represents the primary distortion signal.

Increasing variance indicates growing disagreement between reported financial information and verified financial information.

Repeated high variance may indicate:

- Inaccurate reporting
- Poor verification controls
- Settlement misvaluation
- Behavioral reporting distortion

1.2 Incidence Score Engine

Purpose

Measures frequency of inaccurate submissions.

Equation

$$\text{Incidence} = (\text{Incorrect} / \text{Total}) \times 100$$

R Implementation

```
incidence_engine <- function(incorrect, total){ if(total <= 0) return(0) pmin((incorrect /  
total) * 100,100)  
}
```

Interpretation

Incidence measures recurrence.

Variance measures severity.

Together they identify whether reporting issues are isolated or recurring.

1.3 Consistency Engine

Purpose

Measures behavioral stability.

Equation

$\text{Consistency} = (\text{Successful} / \text{Total}) \times 100$

Interpretation

Low consistency indicates unstable decision-making and inconsistent settlement outcomes.

Persistent inconsistency is a key indicator of negotiator-attributable behavioral risk.

SECTION 2 — NEGOTIATOR ATTRIBUTION LAYER

Objective

Determine whether observed variance originates from operational conditions or from repeated behavioral patterns associated with specific negotiators.

2.1 Source of Variance Predictor

Equation

$\text{SVP} = 0.30(\text{Variance})$

$0.20(\text{Trend})$

0.30(Incidence)

0.20(100 – Consistency)

Interpretation

The predictor estimates the likelihood that variance is structurally embedded within settlement operations.

High scores indicate:

- Repeated inaccuracies
- Increasing variance trends
- Deteriorating reporting consistency

2.2 Negotiator Risk Engine

Equation

Risk=0.40(Incidence)

0.25(Variance)

0.20(Exceptions)

0.15(100 – Accuracy)

Interpretation

The engine attributes risk directly to individual operator behavior.

Elevated scores indicate that specific negotiators are associated with:

- More frequent reporting corrections
- Higher settlement variance
- Greater exception activity
- Reduced reporting accuracy

This engine does not determine intent.

It identifies individuals associated with statistically unusual reporting outcomes.

2.3 Behavioral Variance Detection

Behavioral variance exists when:

Variance ↑ AND Incidence ↑ AND

Consistency ↓

while underlying account characteristics remain materially similar.

This condition suggests operator influence rather than portfolio-level change.

SECTION 3 — SETTLEMENT DISTORTION ANALYSIS

Settlement Percentage Engine

Equation Settlement % = $((\text{Original Debt} - \text{Settlement Amount}) / \text{Original Debt}) \times 100$

Purpose

Measures negotiated debt reduction.

Interpretation

A high settlement percentage is not inherently problematic.

Risk emerges when:

Settlement Reduction ↑

while

Verified Financial Exposure remains unchanged.

This creates potential settlement misvaluation risk.

Settlement Risk Engine

Equation

Settlement Risk = $0.40(100 - \text{Efficiency})$

$0.35(\text{Backlog})$

$0.25(100 - \text{Achievement})$

Interpretation

Identifies situations where settlement outcomes appear favorable while unresolved risk accumulates elsewhere.

SECTION 4 — STRUCTURAL INTEGRITY ASSESSMENT

Liability Engine

Purpose

Measures reliability of reporting entities.

Equation

Liability=

0.35(Accuracy)

0.25(Compliance)

0.30(100 – Incidence)

0.10(Remediation)

Interpretation

Higher liability scores indicate stronger reporting reliability.

Declining scores indicate deteriorating confidence in submitted financial information.

Reporting Integrity Engine

Equation

Integrity=0.30(Accuracy)

0.25(Consistency)

0.25(Compliance)

0.20(100 – Incidence)

Interpretation

Measures trustworthiness of operational reporting.

Reason of Doubt Engine

Equation

Doubt=0.25(Variance)

0.15(Delay)

0.40(Incidence)

0.20(Deviation)

Interpretation

Generates an investigation priority score.

High scores indicate that observed reporting behavior requires additional review.

SECTION 5 — FRAUD AND ESCALATION ANALYTICS

Fraud Score Engine

Equation

Fraud Score=0.35(Variance)

0.30(Incidence)

0.20(Exceptions)

0.15(Triple)

Interpretation

This score does not determine fraud.

Instead, it measures the degree to which observed conditions resemble known risk patterns requiring investigation.

Double Trigger Engine

Activated when:Variance $\geq$ 60

AND

Incidence $\geq$ 60

OR

Settlement Variance Condition Exists

Purpose

Reduce false positives by requiring multiple risk indicators.

Triple Trigger Engine

Activated when:

Variance $\geq$ 60

AND

Incidence $\geq$ 60

AND Legal Action Exists

Purpose

Identify critical escalation scenarios involving simultaneous operational, financial, and legal deterioration.

SECTION 6 — LEGAL LIABILITY EXPOSURE

Litigation Risk Engine

Equation

Litigation Risk = $0.35(\text{Variance})$

$0.35(\text{Incidence})$

$0.30(\text{Settlement Risk})$

Interpretation

Estimates exposure associated with creditor disputes and legal escalation.

Liability Expansion Mechanism

Legal exposure increases when:

1. Reported financial information differs materially from verified information.
2. Settlement decisions rely upon inaccurate information.
3. Creditors discover inconsistencies after settlement completion.
4. Settlements are challenged, reversed, or reopened.

Potential consequences include:

- Recalculated liabilities
- Creditor reclaims

- Settlement invalidation
- Additional client obligations
- Increased client legal exposure

SECTION 7 — POST-SETTLEMENT RE-EXPOSURE LAYER

Broken Settlement Engine

Activation Rule

Settlement Closed

AND

(Reopened OR Disputed OR Legally Reversed)

Result

Broken Settlement Score = 100

Interpretation

A previously resolved account becomes active risk again.

Post-Settlement Income Variance

Equation

Income Variance= $\frac{|(\text{Reported Income} - \text{Verified Income})|}{\text{Verified Income}}$
×100

Purpose

Detect income discrepancies discovered after settlement failure.

Post-Settlement Expense Variance

Equation

Expense Variance= $\frac{|(\text{Reported Expense} - \text{Verified Expense})|}{\text{Verified Expense}}$
×100

Purpose

Detect expense distortions revealed after settlement reversal.

Broken Settlement Trigger

Escalation Logic

If: Broken Settlement = TRUE

AND (Income Variance > 10

OR Expense Variance > 10

OR General Variance > 10

OR Incidence > 30)

Then: Trigger = 1.5

Otherwise: Trigger = 1.25

Interpretation

Settlement failure significantly increases risk weighting throughout the framework.

SECTION 8 — FINANCIAL INTEGRITY INDEX (FII)

Equation

$FII = 0.25(\text{Variance})$

$0.20(100 - \text{Liability})$

$0.15(\text{Doubt})$

$0.20(\text{Negotiator Risk})$

$0.20(\text{Settlement Risk})$

Purpose

Provide a unified governance metric.

Interpretation

Higher FII values indicate greater operational, reporting, and legal risk concentration.

SECTION 9 — GOVERNANCE DECISION ENGINE

Priority Classification

Critical Priority

Legal Action AND (Fraud $\geq$ 80 OR FII $\geq$ 80)

High Priority

Legal Action AND (Fraud $\geq$ 60 OR FII $\geq$ 60)

Elevated

Legal Action Only

Standard

All Other Conditions

SYSTEM FLOW

Reported Financial Data



Variance Detection



Incidence Detection



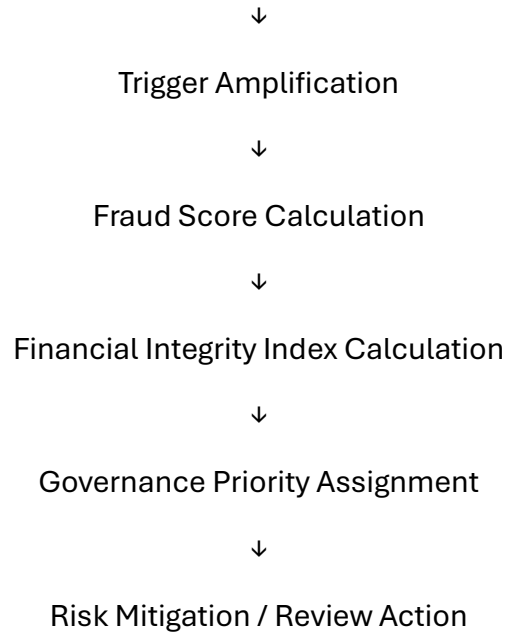
Behavioral Attribution Layer



Settlement Evaluation Layer



Litigation Assessment



1.1 Variance Percentage Engine

Description: Measures deviation between reported and verified financial values.

```
variance_engine <- function(reported, verified){ if(verified <= 0) return(0)  
  abs((reported - verified) / verified) * 100}
```

Example:

- Reported Income: 100,000
- Verified Income: 90,000

During settlement negotiation, a creditor flags the account after discovering the client was **reported as lower risk than verified exposure**, triggering review.

1.2 Incidence Score Engine

Description: Measures frequency of reporting errors.

```
incidence_engine <- function(incorrect, total){ if(total <= 0) return(0)  
  pmin((incorrect / total) * 100, 100)  
}
```

Example:

- 12 incorrect negotiation submissions out of 40 cases
 - Incidence = 30%

Negotiator repeatedly misstates settlement values, causing creditors to question whether **discounts offered reflect true Settlement Value**.

1.3 Consistency Engine

Description: Measures stability of reporting behavior.

```
consistency_engine <- function(successful, total){ if(total <= 0) return(100)
  pmin((successful / total) * 100, 100)}
```

Example:

- 85 consistent reports out of 100
 - Consistency = 85%

Settlement approvals vary widely across similar debt profiles, suggesting **inconsistent negotiation behavior across accounts**.

1.4 Trend Engine

Description: Detects directional drift in reporting.

```
trend_engine <- function(current, previous){pmin(abs(current - previous), 100)}
```

Example:

- Settlement discount last month: 35%
 - This month: 55%
 - Trend spike = 20

Sudden increase in settlement discounts suggests negotiators may be **aggressively reducing debt recovery value without financial justification**.

1.5 4% Trigger Engine

Description: Early warning when variance exceeds threshold.

```
trigger_4pct <- function(variance){ ifelse(variance >= 4, 100, 0)}
```

Example:

- Variance = 6%

Even before settlement is finalized, system flags that **reported exposure is materially understated vs verified debt**, triggering review.

2. LAYER 1 — DETECTION ENGINES

2.1 Source of Variance Predictor

Description: Combines signals to identify emerging discrepancy risk.

```
variance_predictor <- function(variance, trend, incidence, consistency){
```

```
  score <- ( variance * 0.30 + trend * 0.20 + incidence * 0.30 +  
    (100 - consistency) * 0.20 ) pmin(score, 100)}
```

Example:

- Reported debt volatility rising
- Settlement outcomes inconsistent across creditors
- Incidence of negotiation corrections increasing

System detects that **settlement negotiations are becoming structurally unstable across the portfolio**, even before formal disputes arise.

2.2 Variance Acceleration Engine

Description: Measures speed of variance increase.

```
variance_acceleration <- function(v1, v2, v3){ abs((v3 - v2) + (v2 - v1))}
```

Example:

- Variance: 2% → 6% → 14%

Rapid escalation suggests negotiator is progressively **drifting settlement valuations away from verified debt levels.**

2.3 Incidence Momentum Engine

Description: Tracks change in error frequency.

```
incidence_momentum <- function(i1, i2){  
  abs(i2 - i1)  
}
```

Example:

- Incidence: 10 → 35

Negotiation errors increase sharply after new settlement strategy rollout, indicating **process degradation in creditor negotiations.**

3. LAYER 2 — STRUCTURAL ENGINES

3.1 Liability Engine

Description: Measures reporting reliability.

```
liability_engine <- function(accuracy, compliance, incidence, remediation){ score <-  
( accuracy * 0.35 + compliance * 0.25 + (100 - incidence) * 0.30 + remediation * 0.10 )  
  pmin(score, 100)}
```

Example:

- High compliance but rising settlement inaccuracies

Settlement outputs show **systematic misalignment between negotiated and verified income and expense values.**

3.2 Reporting Integrity Engine

Description: Overall data reliability score.

```
integrity_engine <- function(accuracy, consistency, compliance, incidence){ score <- (
```

$$\text{accuracy} * 0.30 + \text{consistency} * 0.25 + \text{compliance} * 0.25 + (100 - \text{incidence}) * 0.) \text{ pmin}(\text{score}, 100)\}$$

Example:

- High accuracy in reporting but inconsistent settlement discounts

Indicates that **financial reporting is clean, but settlement execution is distorting true exposure risk.**

3.3 Reason of Doubt Engine

Description: Flags need for investigation and provides possible context for variance.

```
doubt_engine <- function(variance, delay, incidence, deviation){ score <- ( variance * 0.25
+ delay * 0.15 + incidence * 0.40 + deviation * 0.20 ) pmin(score, 100)}
```

Example:

- Settlement delays increasing
- Variance rising post-negotiation

Suggests negotiators may be **intentionally delaying settlement closure while variance expands in background reporting.**

3.4 Negotiator Risk Engine

Description: Measures operator-driven risk.

```
negotiator_risk <- function(incidence, variance, exception, accuracy){score <- ( incidence
*0.40 + variance * 0.25 + exception * 0.20 + (100 - accuracy) * 0.15 ) pmin(score, 100)}
```

Example:

- One negotiator consistently closes settlements at extreme discounts

System identifies that **a single operator is driving disproportionate debt reduction variance across creditor accounts.**

3.5 Root Cause Engine

Description: Classifies variance origin.

```
root_cause <- function(variance, incidence, exception){  
  if(variance > 60 & incidence > 60) return("Manipulation") if(exception > 70) return("System  
Failure") if(incidence > 50) return("Process Failure") return("Human Error")}
```

Example:

- High variance + high settlement overrides + repeated negotiator adjustments

Root cause classified as **Manipulation due to structured negotiation distortion patterns.**

4. LAYER 3 — SETTLEMENT ENGINES

4.1 Settlement Percentage Engine

Description: Measures discounted debt resolution.

```
settlement_pct <- function(original, settlement){ if(original <= 0) return(0) ((original -  
  settlement) / original) * 100}
```

Example:

- Debt: 10,000
- Settled: 6,000
- Settlement achievement: 40%

Creditor accepts reduced recovery, but system checks if discount is **consistent with verified exposure or artificially inflated variance.**

4.2 Settlement Achievement Engine

Description: Normalized settlement outcome.

```
settlement_achievement <- function(original, settlement){ pmin(settlement_pct(original,  
  settlement), 100)}
```

Example:

- High achievement score but increasing variance elsewhere

Settlement looks successful, but underlying books show **misalignment between negotiated resolution and actual liability risk.**

4.3 Settlement Efficiency Engine

Description: Measures resolution throughput.

```
settlement_efficiency <- function(settled, flagged){ if(flagged <= 0) return(100)
  pmin((settled / flagged) * 100, 100)}
```

Example:

- 200 flagged cases, 80 settled

Backlog suggests negotiators are **closing favorable settlements selectively while leaving high-risk accounts unresolved.**

4.4 Settlement Risk Engine

Description: Measures settlement inefficiency and distortion.

```
settlement_risk <- function(efficiency, achievement, backlog){ score <- (
  (100 - efficiency) * 0.40 + backlog * 0.35 + (100 - achievement) * 0.25 )
  pmin(score, 100)}
```

Example:

- High settlement achievement but large backlog

System flags that **settlements are being skewed toward easy wins while risk-heavy debt remains unresolved.**

4.5 Settlement Variance Trigger

Description: Detects variance occurring during settlement.

```
settlement_trigger <- function(settled, variance){ if(settled & variance > 0) return(100)
  return(0)}
```

Example:

- Settlement closed
- But verified vs reported exposure still differs

Indicates **post-settlement reconciliation mismatch**, often caused by negotiator misvaluation.

5. LITIGATION ENGINES

Legal Action Engine

```
legal_action <- function(flag){
  ifelse(flag == TRUE, 1, 0)
}
```

Example:

Creditor escalates to legal before or after discovering settlement undervalued true exposure.

*Represented as one value until settlement is achieved.

Litigation Risk Engine

```
litigation_risk <- function(legal, variance, incidence, settlement){ if(!legal) return(0) score
  <- (variance * 0.35 + incidence * 0.35 + settlement * 0.30 ) pmin(score, 100)}
```

Example:

Legal action triggered due to **systemic under-reporting of debt prior to settlement execution**.

6. ESCALATION ENGINES

Double Trigger Engine

```
double_trigger <- function(variance, incidence, settlement, legal){ if((variance >= 60 &
  incidence >= 60) |
```

```
(settlement & variance >= 20) | (legal & (variance >= 50 | incidence >=
50))) { return(1.25) } return(1.0)

}
```

Example:

Settlement completed, but variance + legal action combine → system escalates post-facto review.

Triple Trigger Engine

```
triple_trigger <- function(variance, incidence, legal){ if(variance >= 60 & incidence >= 60 &
legal){
return(1.5) } return(1.0)}
```

Example:

Negotiator-driven mispricing leads to litigation + high variance + repeated errors → critical escalation.

Fraud Score Engine

```
fraud_score <- function(variance, incidence, exception, trigger){score <- (
variance * 0.35 +
incidence * 0.30 +
exception * 0.20 +
trigger * 0.15 ) pmin(score, 100)}
```

Example:

Repeated settlement discounts outside verified ranges trigger fraud review of negotiation practices.

7. GOVERNANCE ENGINE

Financial Integrity Index (FII)

```
fii_engine <- function(variance, liability, doubt, negotiator, settlement){ score <- (
    variance * 0.25 +
    (100 - liability) * 0.20 + doubt * 0.15 + negotiator * 0.20 + settlement *
    0.20 ) pmin(score, 100)}
```

Example:

Portfolio shows clean reporting but settlement distortions + negotiator variance pushes FII into high risk band.

Priority Engine

```
priority_engine <- function(fraud, fii, legal){ if(legal & (fraud >= 80 | fii >= 80)) return("Critical
    Priority")
    if(legal & (fraud >= 60 | fii >= 60)) return("High Priority") if(legal)
    return("Elevated") return("Standard")}
```

Example:

Legal action + high fraud score due to settlement misvaluation → immediate escalation to credit committee.

MJ.23.R2

POST-SETTLEMENT FAILURE & RE-EXPOSURE LAYER

This layer activates when a settlement that was previously marked “closed” becomes invalid, reversed, disputed, or legally reopened.

1. BROKEN SETTLEMENT ENGINE

Description

Tracks settlements that fail after closure due to dispute, legal reversal, creditor challenge, or discovery of misvaluation.

```
broken_settlement_engine <- function(settlement_closed, reopened_flag, dispute_flag,
    legal_reversal){ if(settlement_closed == TRUE &&
    (reopened_flag == TRUE || dispute_flag == TRUE || legal_reversal ==
    TRUE)){ return(100)}return(0)}
```

Example

- Settlement marked CLOSED at 60% debt reduction
- 30 days later creditor disputes valuation and reopens case

System flags **post-settlement breakdown risk = 100**, even though original FII was low at closure.

2. POST-SETTLEMENT INCOME VARIANCE ENGINE

Description

Measures variance between reported vs verified income after settlement has been broken.

```
post_settlement_income_variance <- function(reported_income, verified_income,
  broken_flag){ if(broken_flag != TRUE) return(0) if(verified_income <= 0)
  return(0) abs((reported_income - verified_income) / verified_income) * 100}
```

Example

- After settlement break:
 - Reported income used in settlement: 120,000
 - Verified corrected income: 150,000

System detects **income was understated during original settlement negotiation**, contributing to reversal.

3. POST-SETTLEMENT EXPENSE VARIANCE ENGINE

Description

Captures expense misreporting revealed after settlement breakdown.

```
post_settlement_expense_variance <- function(reported_expense, verified_expense,
  broken_flag){ if(broken_flag != TRUE) return(0) if(verified_expense <= 0)
  return(0) abs((reported_expense - verified_expense) / verified_expense) * 100}
```

Example

- Expenses originally overstated to justify deeper settlement discount
- Reported expense: 8,000

- Verified: 5,000

System identifies **inflated expense reporting was used to drive excessive settlement discounting**.

4. BROKEN SETTLEMENT TRIGGER ENGINE

Description

Activates when settlement failure + variance conditions co-occur.

```
broken_settlement_trigger <- function(broken_flag, income_variance, expense_variance,
  variance, incidence){ if(broken_flag == TRUE && (income_variance > 10 |
  expense_variance > 10 | variance > 10 | incidence > 30)){ return(1.5) }
  if(broken_flag == TRUE){ return(1.25) } return(1.0)}
```

Scenario A (moderate escalation)

- Settlement breaks
- Minor income correction (12%)

Trigger = **1.25x escalation**

Scenario B (high escalation)

- Settlement breaks
- Income variance = 25%
- Expense variance = 30%
- Incidence elevated

Trigger = **1.5x escalation (critical re-review)**

5. INTEGRATED VARIANCE + TRIGGER LOGIC LINE (SYSTEM RULE)

This is the **formal governance rule hook** that ties into your existing engine stack.

IF Settlement_Status == "BROKEN"

AND (Income_Variance > 0 OR Expense_Variance > 0 OR General_Variance > Threshold OR Incidence_Score > Threshold)

THEN:

Activate Broken_Settlement_Engine

Apply Broken_Settlement_Trigger

Elevate Fraud_Score Weighting

Recalculate FII

Flag Account = "Post-Settlement Re-Exposure Risk"

6. GOVERNANCE IMPACT (HOW THIS CHANGES THE SYSTEM)

This **fourth phase in lifecycle intelligence**:

BEFORE:

Detection → Structuring → Settlement → **Post-Settlement Integrity Layer** → **Governance**

WHAT THIS SOLVES

This module specifically detects:

Hidden Financial misrepresentation in debt settlement negotiations and associate risk even if revealed after closure

Settlements reversed due to inaccurate income/expense reporting

Negotiator-driven distortion that only becomes visible post-event

“False hardship” accounts

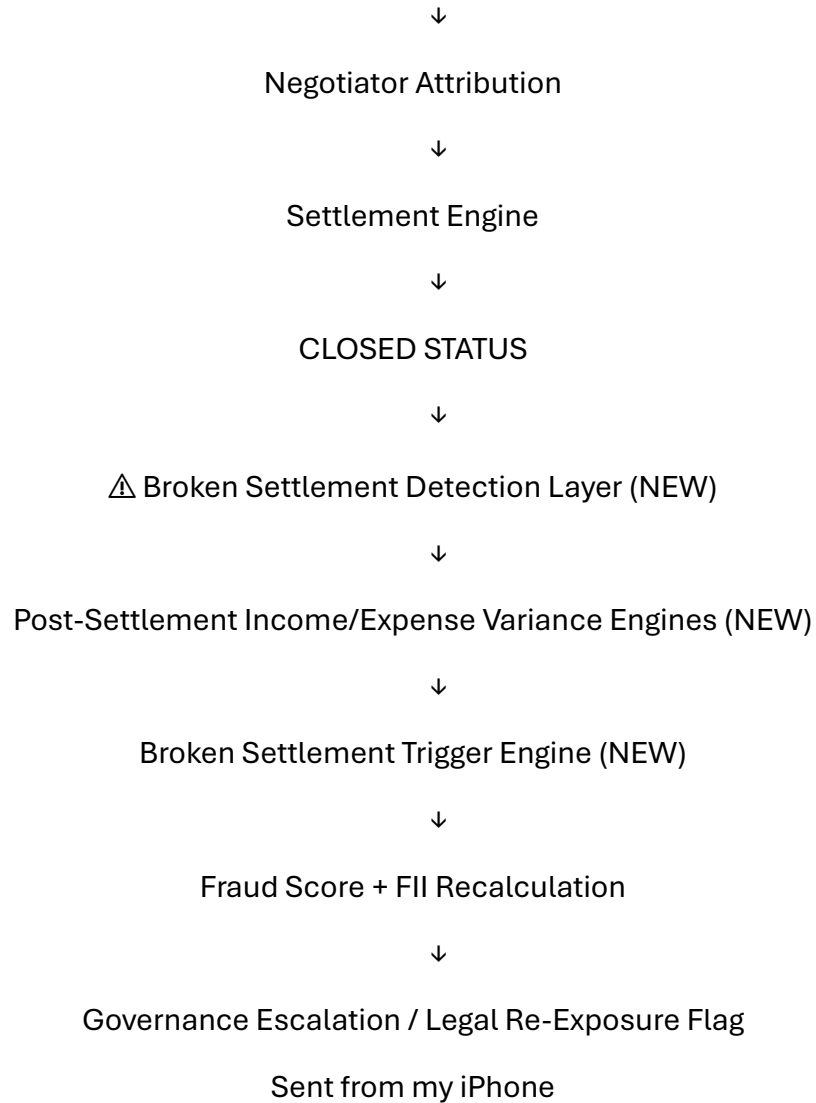
Legal reactivation risk after settlement

8. SYSTEM FLOW

Reported Financials

↓

Variance + Incidence Detection



MJ.23.R2 — GLOSSARY OF TERMS

A

Accuracy Score

Measure of correctness between reported and verified financial data.

→ Helps identify unreliable reporting sources.

Audit Trail Integrity

Degree to which financial actions can be traced end-to-end.

→ Ensures forensic transparency in settlement disputes.

B

Behavioral Variance

Deviation caused by human actions (especially negotiators).

→ Helps separate system error from operator-driven distortion.

Broken Settlement

A previously closed settlement that is later reopened, disputed, or legally reversed.

→ Indicates hidden post-settlement risk exposure.

C

Compliance Score

Degree of adherence to financial reporting rules and procedures.

→ Ensures regulatory alignment.

Consistency Score

Stability of reporting behavior across time.

→ Detects erratic negotiator patterns.

D

Debt Adjustment Ratio

Difference between original debt and negotiated settlement outcome.

→ Measures aggressiveness of settlement behavior.

Delay Score

Measures lateness in reporting or resolving financial issues.

→ High delay correlates with hidden risk accumulation.

E

Exception Score

Count of unusual or non-standard financial adjustments.

→ Flags abnormal negotiation behavior.

Escalation Threshold

Defined point where triggers activate fraud or review processes.

→ Controls system sensitivity.

F

Financial Integrity Index (FII)

Composite score representing overall financial reliability.

→ Primary governance output.

Fraud Score

Likelihood that observed behavior reflects intentional manipulation.

→ Used for investigative prioritization.

G**Governance Priority Level**

Classification of urgency (Standard, High, Critical).

→ Drives audit and legal response.

H**Historical Settlement Drift**

Gradual degradation in settlement accuracy over time.

→ Detects systemic negotiation bias.

I**Incidence Score**

Percentage of incorrect or revised financial submissions.

→ Core behavioral risk indicator.

Income Variance

Difference between reported and verified income.

→ Key driver of settlement distortion.

J**Judgement Bias (Negotiator)**

Systematic distortion introduced by decision-makers in settlements.

→ Core MJ.23.R2 behavioral risk factor.

K**Key Exposure Event**

Any settlement or variance event above threshold significance.

→ Used for escalation triggers.

L

Liability Score

Reliability rating of reporting entity.

→ High score = trustworthy source.

Legal Reversal Event

Formal legal action that invalidates a settlement.

→ Activates broken settlement engine.

M**Misreporting Index**

Combined measure of inaccurate financial submissions.

→ Aggregates incidence + variance.

N**Negotiator Risk Score**

Risk attributed to individual negotiator behavior.

→ Core attribution engine output.

O**Operational Drift**

Gradual divergence from expected financial behavior patterns.

→ Early indicator of systemic risk.

P**Post-Settlement Variance**

Difference discovered after settlement closure.

→ Reveals hidden misvaluation.

Priority Engine Output

Classification of urgency for review.

→ Standard / High / Critical.

Q**Quality Degradation Score**

Measure of declining reporting or negotiation quality.

→ Tracks long-term system health.

R

Reason of Doubt Score

Measure of uncertainty requiring investigation.

→ Used in fraud escalation logic.

Reporting Source Reliability

Trust level of data origin entity.

→ Used in liability engine.

S**Settlement Efficiency**

Ratio of resolved settlements to flagged cases.

→ Measures operational effectiveness.

Settlement Closure Status

Indicates whether a settlement is marked complete.

→ Can later be invalidated by broken settlement engine.

T**Trigger Amplification Factor**

Multiplier applied when multiple risk conditions align.

→ Increases fraud/FII sensitivity.

Triple Trigger Event

Simultaneous activation of variance, incidence, and legal escalation.

→ Critical system failure indicator.

U**Underreporting Index**

Measure of systematically understated financial values.

→ Often negotiator-driven.

V**Variance Percentage**

Core metric comparing reported vs verified values.

→ Primary financial distortion signal.

Variance Acceleration

Rate of change in variance over time.

→ Detects worsening financial distortion trends.

W

Weighted Risk Score

Composite risk score across multiple engines.

→ Used in FII calculation.

X

X-Factor Distortion (Negotiation Effect)

Unexplained deviation introduced during negotiation process.

→ Captures intangible behavioral bias.

Yield Adjustment (Settlement Yield)

Final effective recovery percentage after settlement.

→ Measures creditor outcome effectiveness.

Zero Integrity Event

Severe case where multiple engines indicate systemic failure.

→ Equivalent to enterprise-wide breakdown condition.

Broken Settlement Engine

Detects post-closure invalidation of settlements.

→ Converts “closed” cases back into active risk.

Double Trigger System

Activates when variance + incidence thresholds are jointly exceeded.

→ Prevents false positives from isolated signals.

Fraud Score Engine

Combines financial + behavioral + structural risk.

→ Drives investigative prioritization.

Financial Integrity Index (FII)

Final governance score.

→ Single metric for enterprise financial health.

